Physics
Higher level
Markscheme
B.4 Thermodynamics

8 November 2026

W4ss Academy practice test 3 | **Topic** B.4 Thermodynamics | **Ref.** W4SS–PHY–B4–03

Student name

Paper 1A and Paper 2

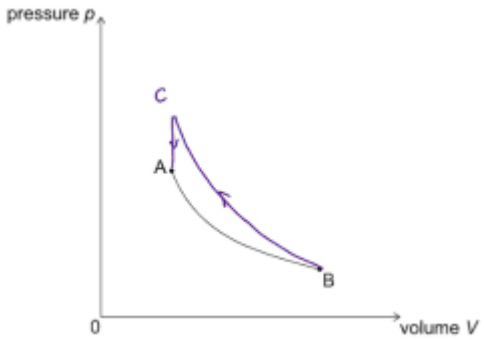
Instructions to students

- This markscheme is for teachers and for self-assessment after the test.
- Paper 1A: one mark per correct answer. The method (in French) shows how to find each answer.
- Paper 2: award one mark per ✓ in the markscheme. Accept equivalent answers (OWTTE).
- Total: **[20 marks]**.

Paper 1A — answers and methods

Question	Answer	Method
1.	**D**	1. Le deuxième principe s'applique au système **et** à son environnement : leur entropie totale ne diminue jamais. 2. L'entropie d'un système non isolé peut donc diminuer, mais seulement si celle de l'environnement augmente au moins autant.
2.	**D**	1. Deuxième principe : l'entropie de l'Univers augmente lors d'une transformation irréversible et reste constante lors d'une transformation réversible. 2. Elle ne diminue donc jamais. « Augmente toujours » (A) est faux pour une transformation réversible.
3.	**D**	1. Le réfrigérateur consomme de l'énergie électrique, finalement dissipée en énergie thermique dans la pièce fermée. 2. Ces transformations sont irréversibles : l'entropie de la pièce augmente (le réfrigérateur ne la refroidit pas, porte ouverte).

Paper 2 — markscheme

Question	Marks	Answers
1. (a)	**[2]**	use of pV = constant ✓ P_B = 43 «Kpa» ✓ ***Award [2]*** *for* ***BCA***
1. (b)	**[2]**	concave curved line from B to locate C with a higher pressure than A ✓ vertical line joining C to A ✓ *Allow* ***ECF*** *from* ***MP1*** *i.e.,* ***award [1]*** *for first process locating C at a lower pressure than A, then vertical line to A.* *Arrows on the processes are not needed.* *Point C need not be labelled.* 
1. (c)	**[2]**	***ALTERNATIVE 1*** use of $TV^{\frac{2}{3}}$ = constant «so 300 ($3V_A$) $^{\frac{2}{3}}$ = T_C (V_A) $^{\frac{2}{3}}$ » ✓ T_C = 624 «K» ***OR*** T_C = 351 «°C» ✓ ***ALTERNATIVE 2*** use of $pV^{\frac{5}{3}}$ to get either p_c = 43 (3) $^{\frac{5}{3}}$ ***OR*** p_c = 268 «kPa» ✓ « T_C = 268 × 300/129 = so » T_C = 624 «K» ***OR*** T_C = 351 «°C» ✓
1. (d)	**[1]**	***ALTERNATIVE 1*** «the process is adiabatic so» ΔQ = 0 ✓ ***ALTERNATIVE 2*** The compression is reversible «so ΔS = 0» ✓ ***OWTTE***
1. (e)	**[1]**	area under curve AB is less than area under curve BC ✓ *Do* ***not*** *allow* ***ECF*** *from part* ***(b)***

1. (f)	**[2]**	«W = 0 so» Q = ΔU ✓ «$\Delta U = \frac{3}{2} \times 53.2 \times R \times (351 - 27)$ so » $\Delta U = 2.15 \times 10^{5}$ «J» ✓ ***Award [2]*** *for* ***BCA***
2. (a)	**[2]**	6 microstates ✓ Any of the six particles can be the one of the highest energy ✓
2. (b)	**[2]**	$k_B \ln 120 - k_B \ln 6$ ✓ $3.0k_B$ ✓
2. (c)	**[3]**	The second law predicts that isolated systems spontaneously evolve towards high-entropy states ✓ From (c), the entropy of B is greater than that of A ✓ The final state will likely be similar to B / contain relatively many low-energy particles «of different energies» ✓